RP3-H

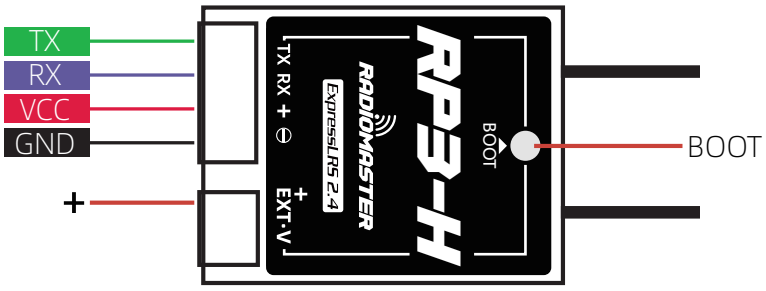
Designed specifically for use with our Nexus Helicopter Flybarless control unit and compatible with all CRSF/S.BUS enabled flight controllers the RP3-H ExpressLRS receiver comes with a built-in TCXO and an integrated housing design that makes the receiver tougher and ready to mount in modern helicopter frames. With the RP3-H receiver, you can get reliable, stable, and accurate signal reception, which enhances your flight experience and allows for superior performance even in demanding environments. The RP3-H also features an Ext-V telemetry input for compatibility with older ESCs that do not provide battery voltage telemetry via the control unit.

The RP3-H receiver uses the Skyworks SKY66112 power amplifier too increase telemetry performance, the PA provides up to 100mw of telemetry power, and antenna switching for antenna diversity. It is developed around the ESP8285 MCU and SX1281IMLTRT RF Chip, offering Lora and FLRC 2.4GHz RF links that provide incredible range and low latency. With Wifi built-in, firmware updates can be done on your craft without the need to remove the receiver.

TCXO (temperature compensated crystal oscillator)
The TCXO ensures high stability and accuracy in frequency control, compensating for temperature variations. This results in consistent and reliable signal reception, even in extreme temperature conditions. It also reduces frequency drift over time, providing long-term frequency stability and preventing signal degradation. Click https://www.expresslrs.org/hardware/crystal-frequency-error to learn more about the TCXO.

RP3-H Specifications
- Item: RP3-H ExpressLRS 2.4GHz Nano Receiver
- Type: ISM
- MCU: ESP8285
- RF Chip: SX1281
- Telemetry RF power: Max 100mW
- Antenna: 150mm 2.4GHz Antenna x2
- Frequency Range: 2.404 - 2.479 GHz
- Maximum receive refresh rate: 500Hz / F1000Hz
- Minimum receiver refresh rate: 25Hz
- Working voltage: 5v
- Battery voltage detection range: DC 4.0 - 35V
- Weight: 5.1g (Including two antenna)
- Dimension: 28mm*24mm*8mm
- Firmware Version: ExpressLRS v3.3.1 pre-installed
- FW Target: RadioMaster RP3-H Diversity 2400 RX
- Bus interface: CRSF/S.Bus

RP3-H 技术参数
- 名称：RP3-H ExpressLRS 2.4GHz Nano 接收机
- 类型：ISM
- 单片机：ESP8285
- 射频芯片：SX1281
- 射频功率: 最大100mW
- 天线：2根 150mm 2.4GHz 天线
- 频率范围：2.404 - 2.479 GHz
- 最大接收刷新率：500Hz/F1000Hz
- 最低接收刷新率：25Hz
- 工作电压：DC 4.5 - 8.4V
- 电压检测范围： DC 4.0 - 35V
- 重量：5.1g （含2个天线）
- 尺寸： 28mm*24mm*8mm
- 固件版本：预装 ExpressLRS v3.3.1
- 固件目标：RadioMaster RP3-H Diversity 2400 RX
- 总线接口：CRSF/S.Bus



对频方法

1：关闭遥控器
2：重复给接收机上电三次，接收机灯双闪，表明接收机处于对频模式
3：开启遥控器，进入ExpressLRS的LUA操作界面，选择到【BIND】，确认
4：接收机灯常亮表明对频成功

Traditional Binding
For traditional binding, the binding phrase must be commented out in user_defines on the RX.

1: Power off your transmitter
2: Plug in and unplug your receiver three times
3: Make sure the LED is doing a quick double blink, which indicates the receiver is in bind mode
4: Use the [BIND] button on the ExpressLRS Lua script, which sends out a binding pulse
5: If the receiver has a solid light, it's bound!

	灯状态 Status light	工作状态 Meaning
接收机断开 Receiver disconnected	500毫秒的周期慢闪 Slow flash (500ms)	等待连接遥控器 Waiting for transmitter
WIFI升级模式 WIFI upgrade mode	25毫秒的周期快闪 Fast flash (25ms)	WIFI功能开启 WIFI On
对频模式 Bind mode	双闪 Double flash	接收机进入对频模式 Receiver in bind mode
模式匹配失败 Mode matching failed	三闪 Triple flash	接收机连接正常，但是工作模式不匹配 Receiver functional, mode selection error
连接状态 Receiver connected	常亮 Solid	接收机正常工作 Receiver functional

For more information, please visit the ELRS website:
更多操作细节，请参考ELRS官方网站： https://www.expresslrs.org/2.0/